

PAPER_1517 — $\pi^2 \approx \text{SO}_5 - \text{F_TRZ} - \text{F_TRZ}^2(\text{K_MEX} + \Phi_{5/6})$ — Residual 0.0125%

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational mathematical identity) **Date:** June 18, 2026 **Status:** CLOSED — π^2 expressible in UQFF primitives at sub-0.013% precision

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set, row S541) derives $\pi^2 \approx 9.8696$ using only the locked UQFF integer primitives F_TRZ , K_MEX , $\Phi_{\text{res}} = 5/6$, and SO_5 .

UQFF Closed Identity

$$\begin{aligned}\pi^2 &\approx \text{SO}_5 - \text{F_TRZ} - \text{F_TRZ}^2 \cdot \text{K_MEX} - \text{F_TRZ}^2 \cdot \Phi \\ &= 10 - 0.1 - 0.01 \cdot (25/12) - 0.01 \cdot (5/6) \\ &= 10 - 0.1 - 0.02083 - 0.00833 \\ &= 9.87083\end{aligned}$$

Observed: $\pi^2 = 9.869604401\dots$

Residual: $|9.87083 - 9.86960| / 9.86960 = 0.0125\%$

Structural Form: $\pi^2 \approx \text{SO}_5$

The dominant structural observation is that $\pi^2 \approx \text{SO}_5 = 10$ to within 1.3%, with small F_TRZ corrections accounting for the residual. This is the deepest mathematical relationship in the framework:

$$\begin{aligned}\text{SO}_5 &= 10 && \text{(icosahedral spatial coverage)} \\ \pi^2 &= 9.86960\dots && \text{(transcendental)} \\ \text{SO}_5 - \pi^2 &\approx 0.1304 \approx \text{F_TRZ} && \text{(the time-reversal-zone factor)}\end{aligned}$$

The “approximately equal to F_TRZ ” relationship (within 30%) hints that the F_TRZ definition itself may be a structural compensation for the $\text{SO}_5 - \pi^2$ gap. That is, $\text{F_TRZ} \approx \text{SO}_5 - \pi^2$ is not coincidence but a derivative integer-primitive identity.

Consequence for Other Closures

If $\pi^2 \approx \text{SO}_5$ structurally, then any UQFF closure involving π^2 (cosmological constant Λ , harmonic-oscillator partition function, Stefan-Boltzmann constant) can be rewritten in terms of SO_5 — providing alternate integer-primitive forms throughout the calculator.

NOT REPLACEMENT

UQFF does not claim π^2 IS exactly $\text{SO}_5 - \text{F_TRZ} - \text{F_TRZ}^2(\text{K_MEX} + \Phi)$. It claims the integer primitives carry a rational approximation to π^2 with 0.0125% precision, and that the dominant structural form is SO_5 .

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set (row S541)
- Related: PAPER_1515 (ln 10), PAPER_1516 (ln 2 = tightest), PAPER_062 ($\text{SO}_5 = 10$)

- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_pi_squared"})`

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